Measures of Disease Occurrence - Part 1

APHS 20203 • Epidemiology & Biostatistics

Course Learning objectives

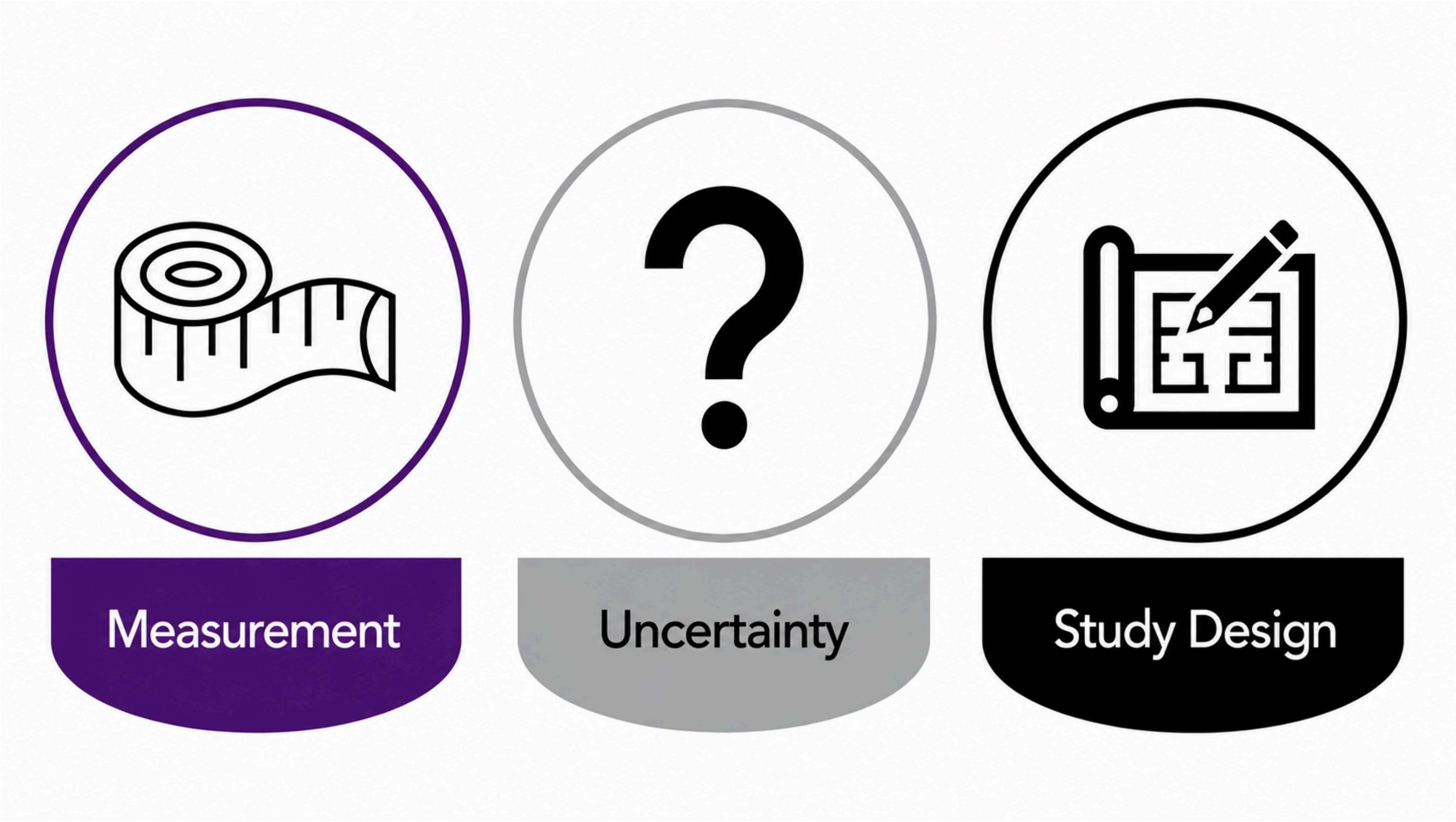
1. Define epidemiology and key associated terminology, especially confounding.
2. Distinguish among association, prediction, and causation.
3. Explain the Rothman model of sufficient-component causes (RMSCC) and the concept of multifactorial causation.
4. Explain the limitations of anecdotal evidence in drawing epidemiologic conclusions.
5. Explain why disseminating risk information is sometimes necessary but rarely sufficient to improve population health.

Lecture Learning objectives

By the end of this lesson, you'll be able to:

1. Define counts and rates.
2. Distinguish crude, specific, adjusted rates.
3. Compute and interpret prevalence and incidence.

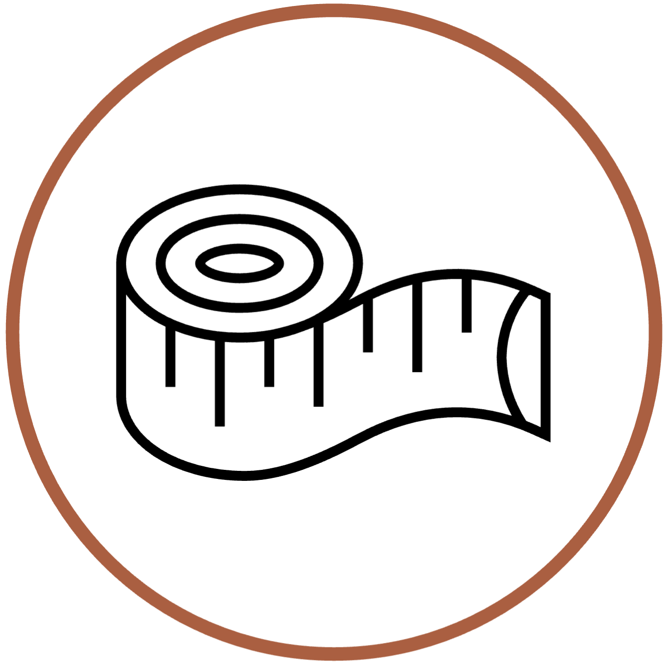
Measurement, Uncertainty, and Study Design



What is Epidemiology?

> Epidemiology is concerned with the distribution and determinants of health and diseases, morbidity, injuries, disability, and mortality in populations. Epidemiologic studies are applied to the control of health problems in populations (Quinlan 2025).

Measurement Makes Characteristics Observable

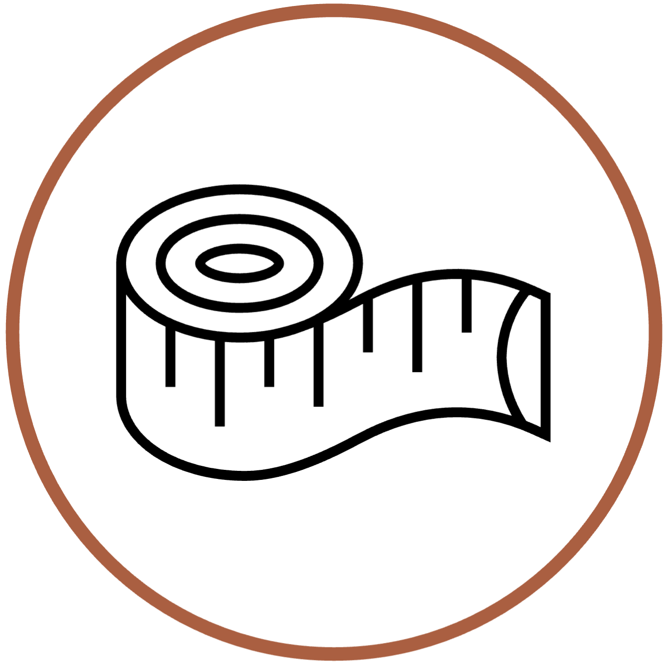


Measurement

Epidemiology is concerned with the **distribution** and determinants of health and diseases, morbidity, injuries, disability, and mortality in populations. Epidemiologic studies are applied to the control of health problems in populations (Quinlan 2025).

- Observe and assign values to relevant characteristics.
- Look for patterns among those values.

What Does It Mean to Observe?



Measurement

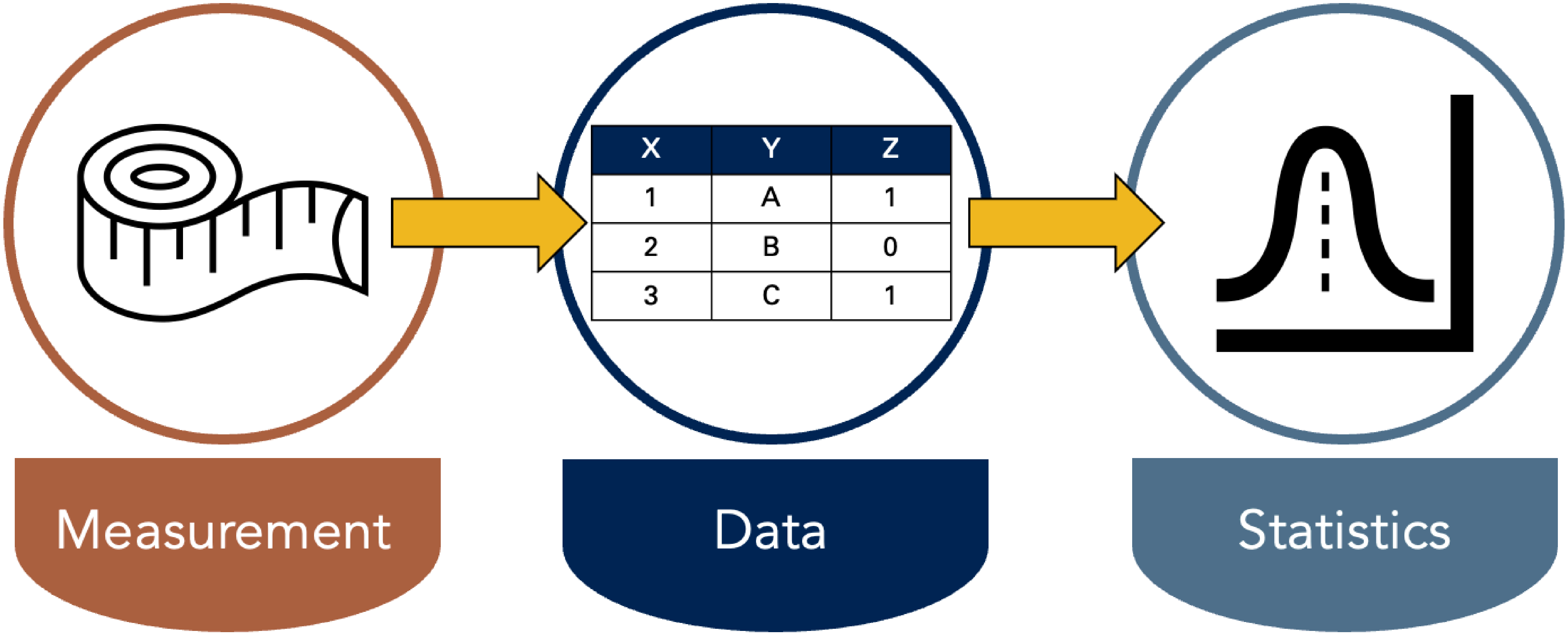
> Epidemiology is concerned with the distribution and **determinants** of health and diseases, morbidity, injuries, disability, and mortality in populations. Epidemiologic studies are applied to the control of health problems in populations (Quinlan 2025).

- Observe and assign values to relevant characteristics.
- Look for patterns among those values.

OBSERVE

"To be aware of" or "have knowledge about."

Measurement Produces Data; Statistics Finds Patterns



Description, Prediction, and Causation

DESCRIPTION	PREDICTION	CAUSATION

Description Does Not Necessarily Seek Associations

- Description examines distributions of single variables: middle, spread, shape, count, and proportion.
- It supports resource management and planning.
- Examples:
  - How many ventilators are available in Texas?
  - What is the average age of people living in Florida?
  - How much time elapses, on average, between exposure to a pathogen and the occurrence of disease symptoms?

Description Can Also Compare Distributions

- Sometimes description does look for associations.
- It may compare distributions of single variables within levels of another variable.
- Examples:
  - Are more ventilators available in Texas or New York?
  - Are people older, on average, in Florida or Pennsylvania?
  - Is symptom onset quicker, on average, for cholera or *E. coli*?

Measures of Occurrence Support Description

- Counts
- Incidence
- Prevalence
- Odds

These measures can be useful on their own and for hypothesis generation.

Sources of Uncertainty: Few Checklists

1. Few checklists

Sources of Uncertainty: Add Statistical Uncertainty

1. Few checklists
2. Statistical uncertainty

Sources of Uncertainty: Add Causal Uncertainty

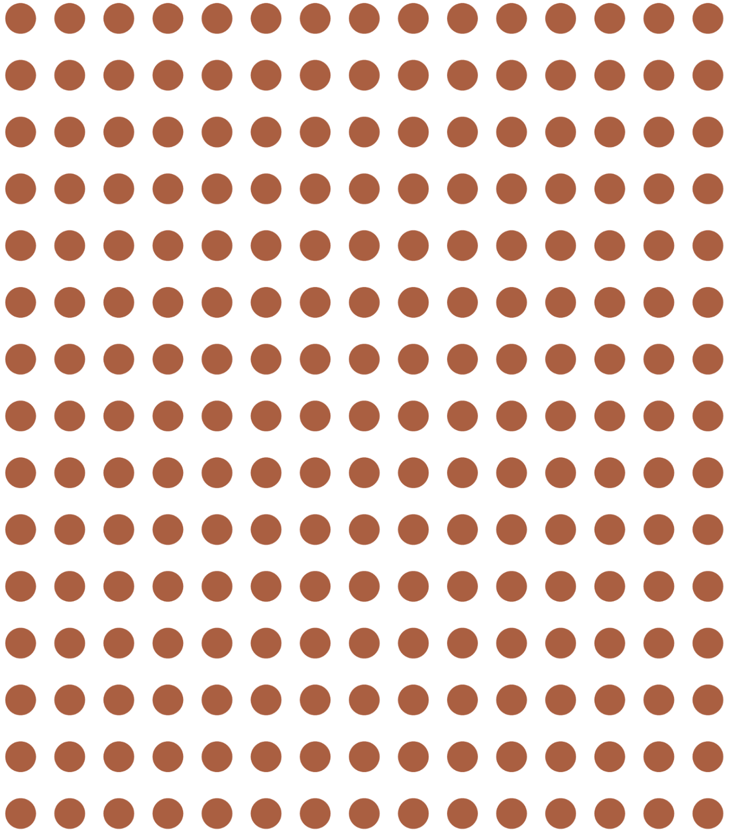
1. Few checklists
2. Statistical uncertainty
3. Causal uncertainty

Occurrence Is Defined Within Populations

> Epidemiology is concerned with the distribution and determinants of health and diseases, morbidity, injuries, disability, and mortality in **populations**. Epidemiologic studies are applied to the control of health problems in populations (Quinlan 2025).

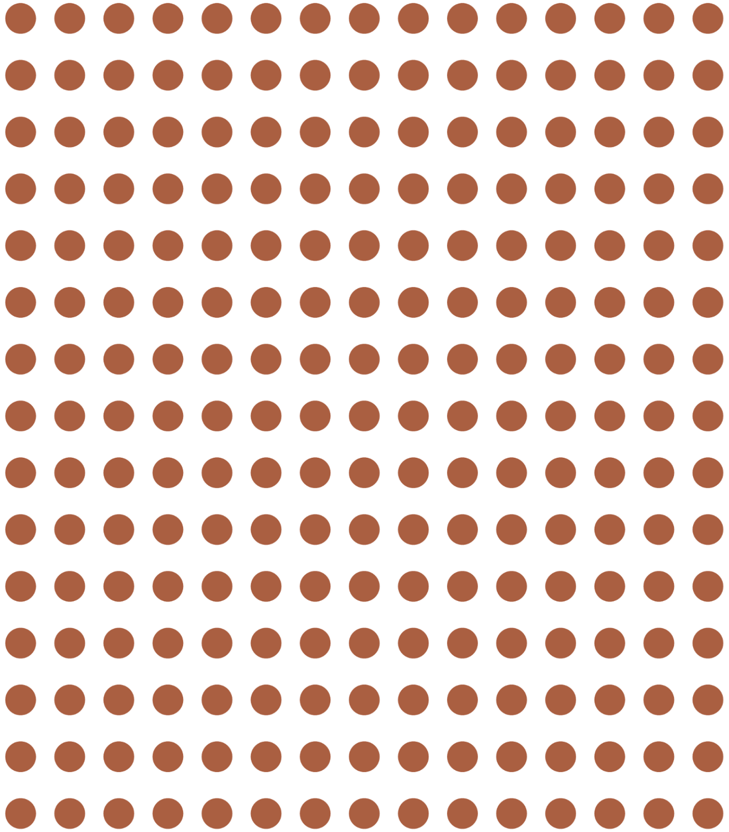
Population

> The simplest definition of a **population** is a group of people who share characteristics or meet criteria that define membership in the population (Lash et al. 2021).

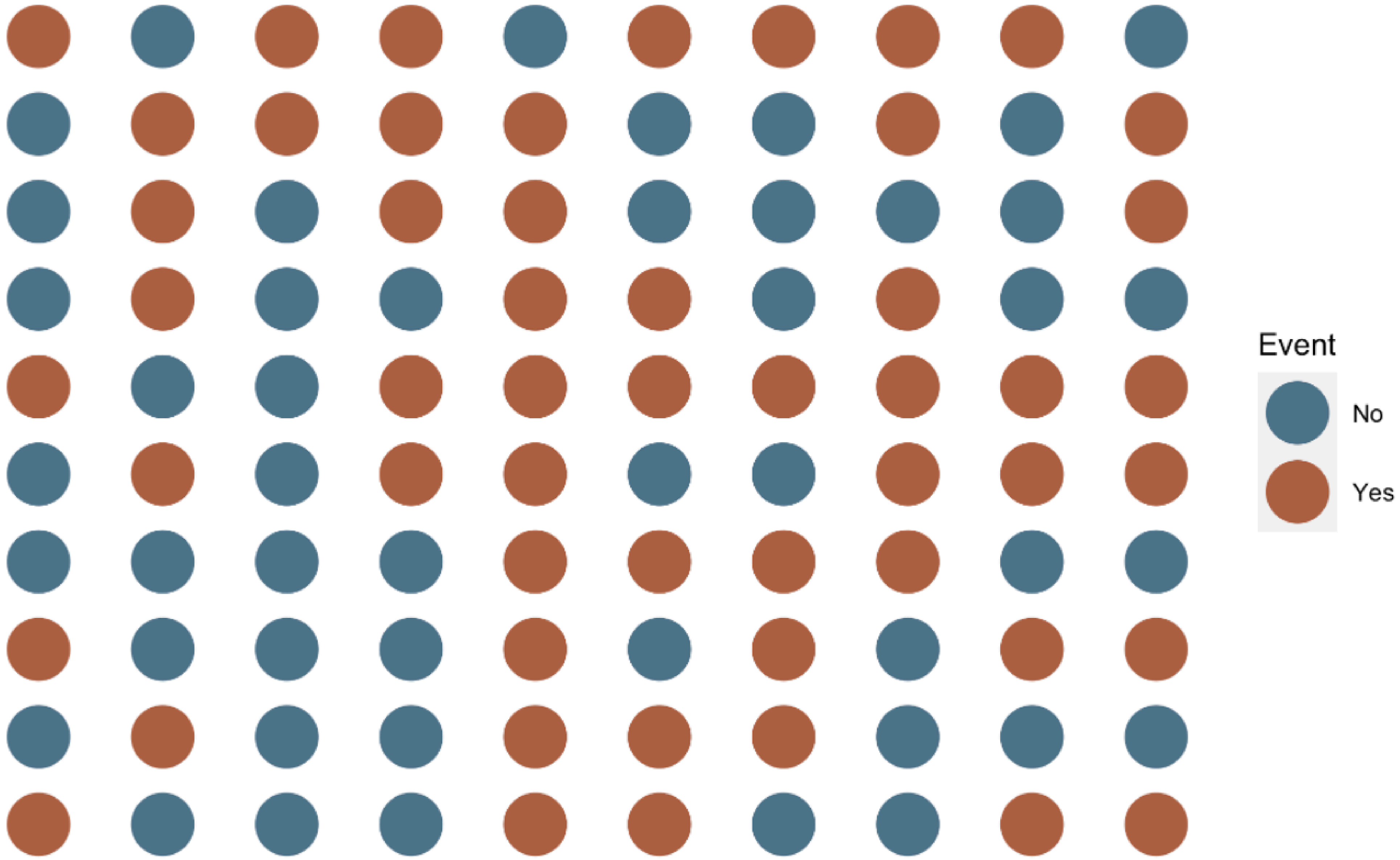


A Population Also Has a Time Period

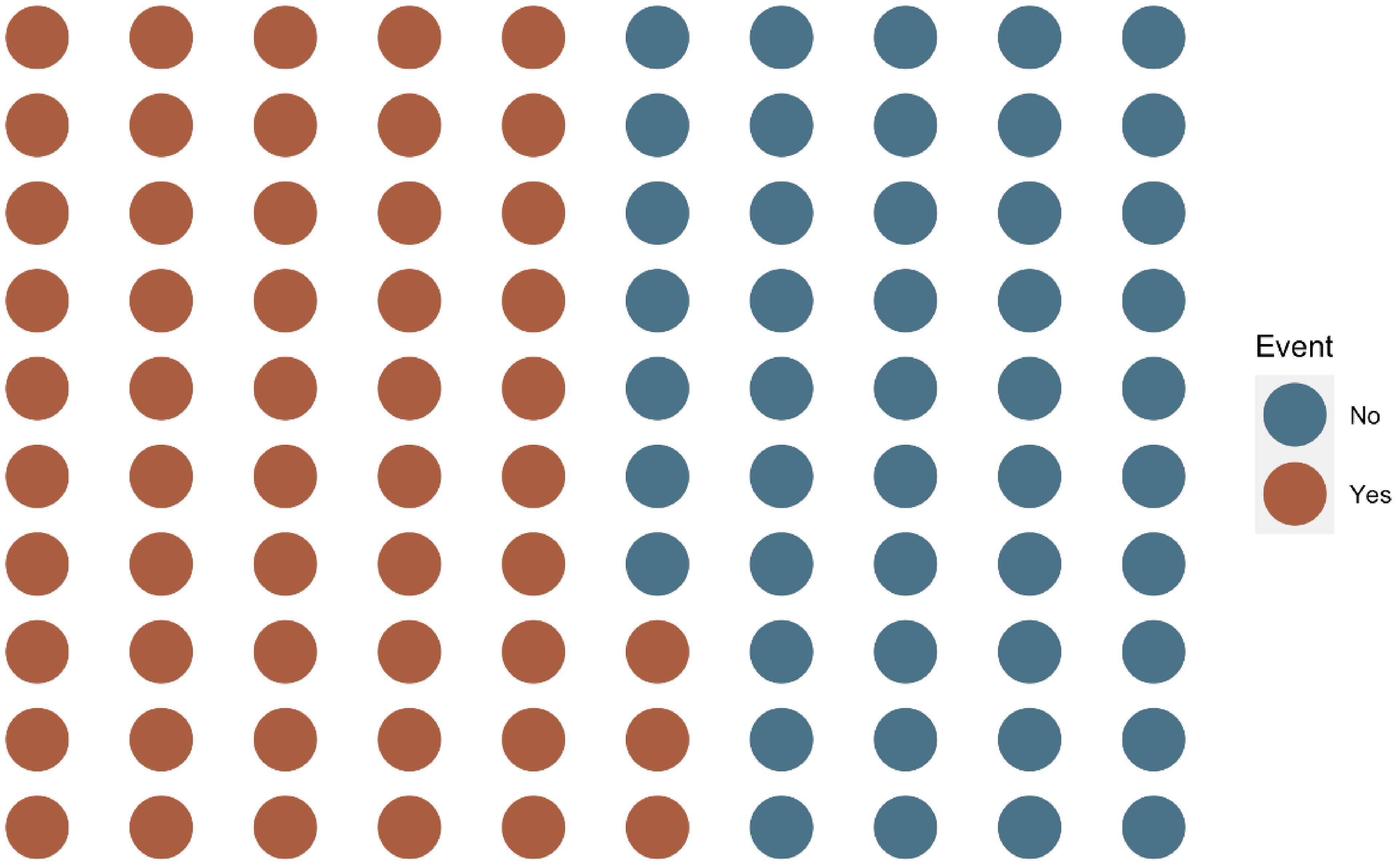
The simplest definition of a **population** is a group of people who share characteristics or meet criteria that define membership in the population [**during a defined time period**] (Lash et al. 2021).



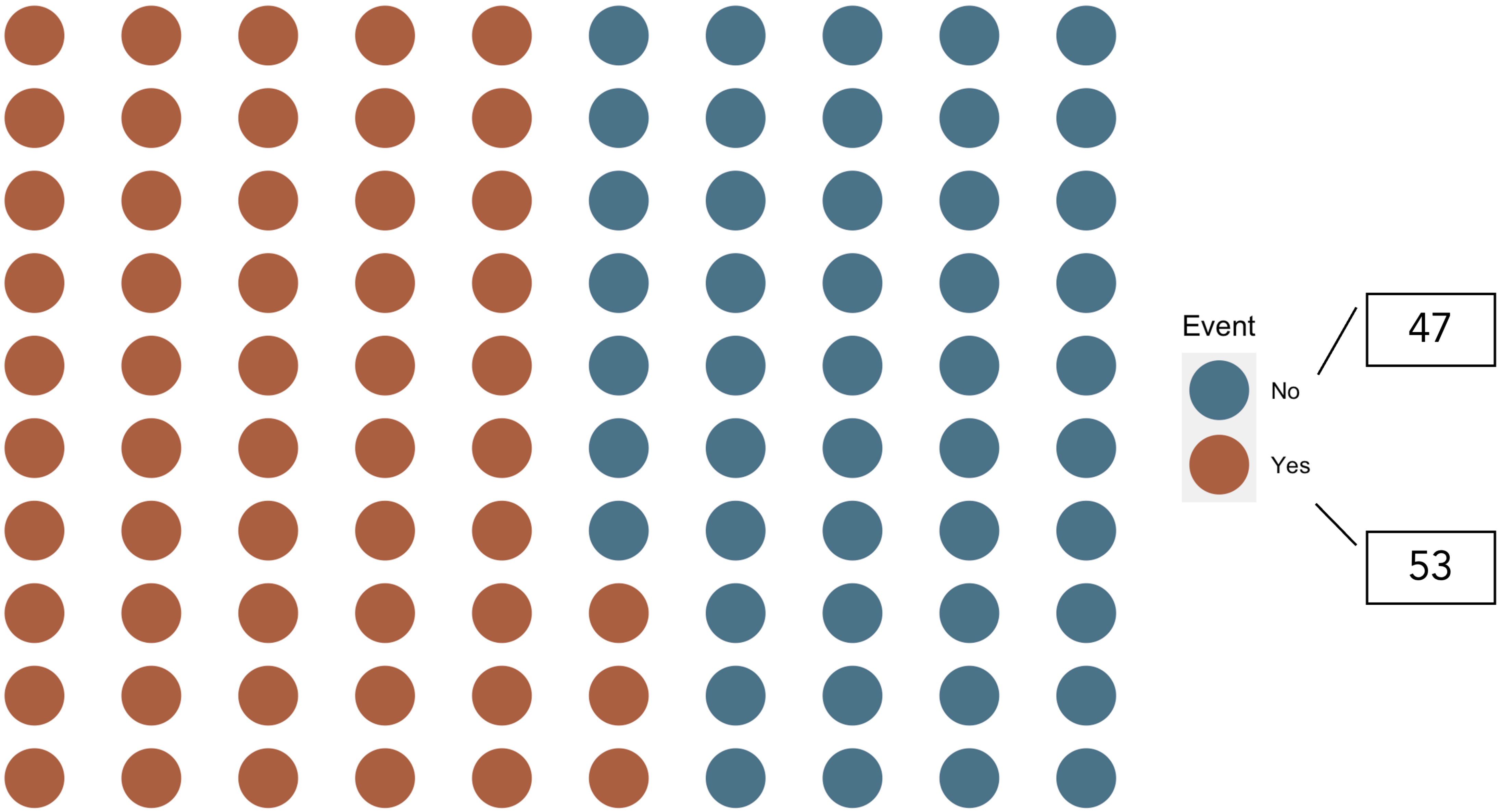
Counts (n): Events in a Population



Counts (n): Group the Event Categories



Counts (n): Reveal the Totals



Prevalence Describes Existing Conditions

- Prevalence provides information about the presence or absence of a condition of interest, such as a disease, exposure, or characteristic, in a population.
- It describes the extent to which the condition is present in a population during a given time frame.
- Asking about the prevalence of diabetes means asking how many, or more likely what proportion, of people currently living in the population have diabetes.

Source: (Westreich 2019)

Prevalence Count

- A count of people in a population with the condition of interest during a given time frame.
- As few as 0 people and as many as all people can be living with the condition.
- Range: 0 to infinity.

Source: (Westreich 2019)

Prevalence Proportion

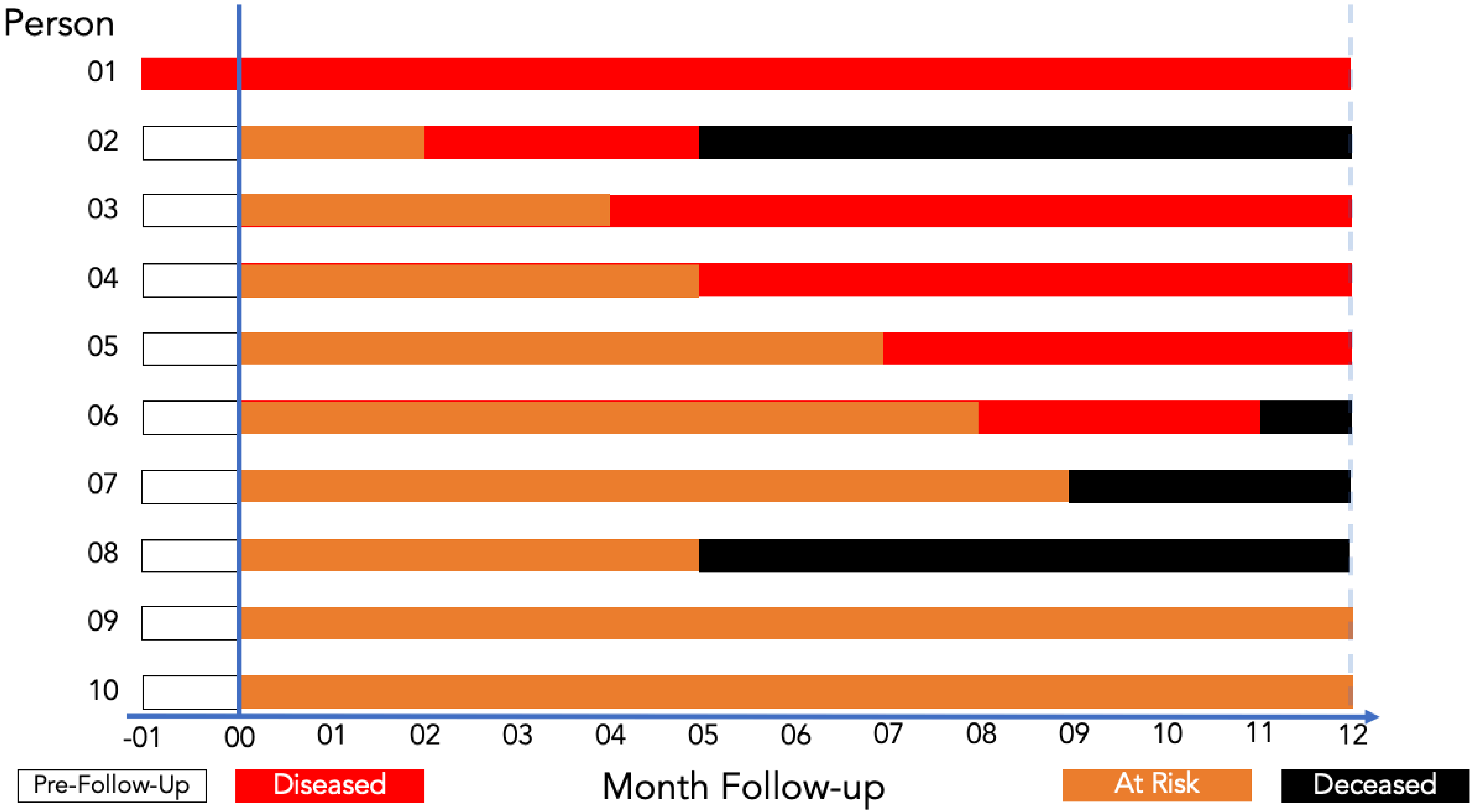
- The proportion or percentage of the population that presently has the condition of interest or presently has a history of the condition during a specified period, such as a history of cancer in the past 10 years.
- Since none, some, or all population members can have the condition or a history of it, prevalence proportion ranges from 0 to 1, much like a probability.

Source: (Westreich 2019)

Prevalence Proportion: Formula

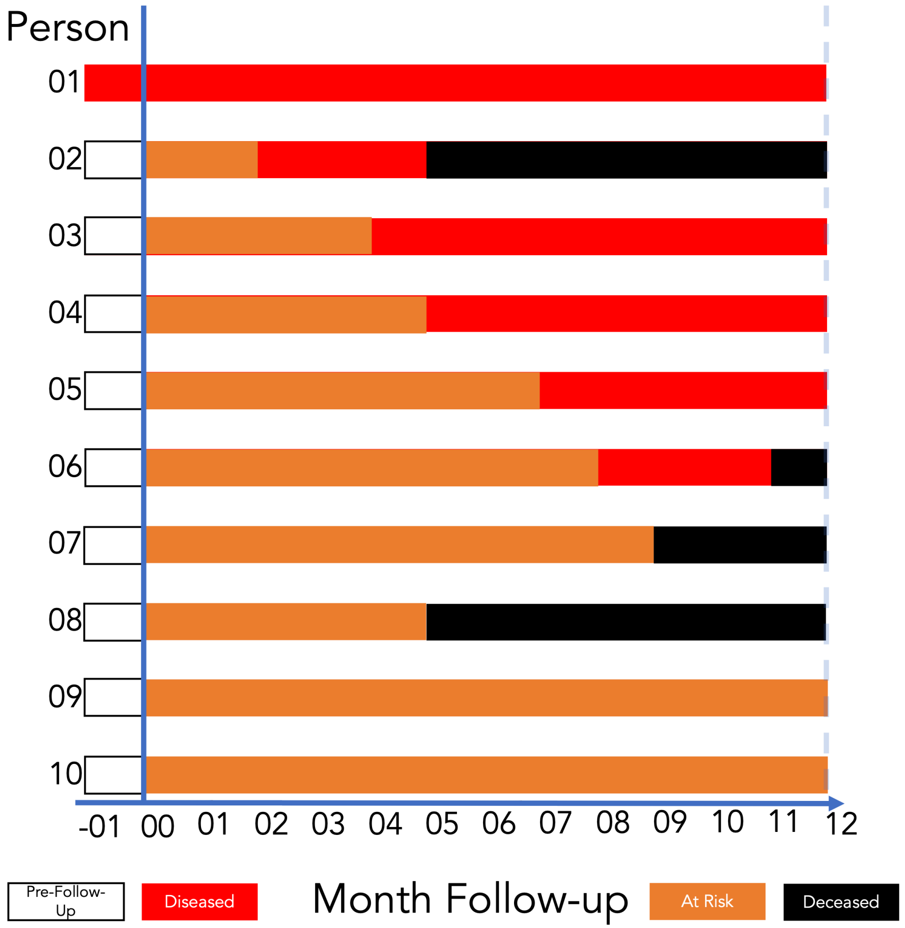
$$\text{Prevalence proportion} = \frac{\text{Count with the condition}}{\text{Living members of the population}}$$

Follow-Up Experience for 10 People



Source: Lash TL, VanderWeel TJ, Haneuse S, Rothman KJ. *Modern Epidemiology*. fourth. Wolters Kluwer; 2021.

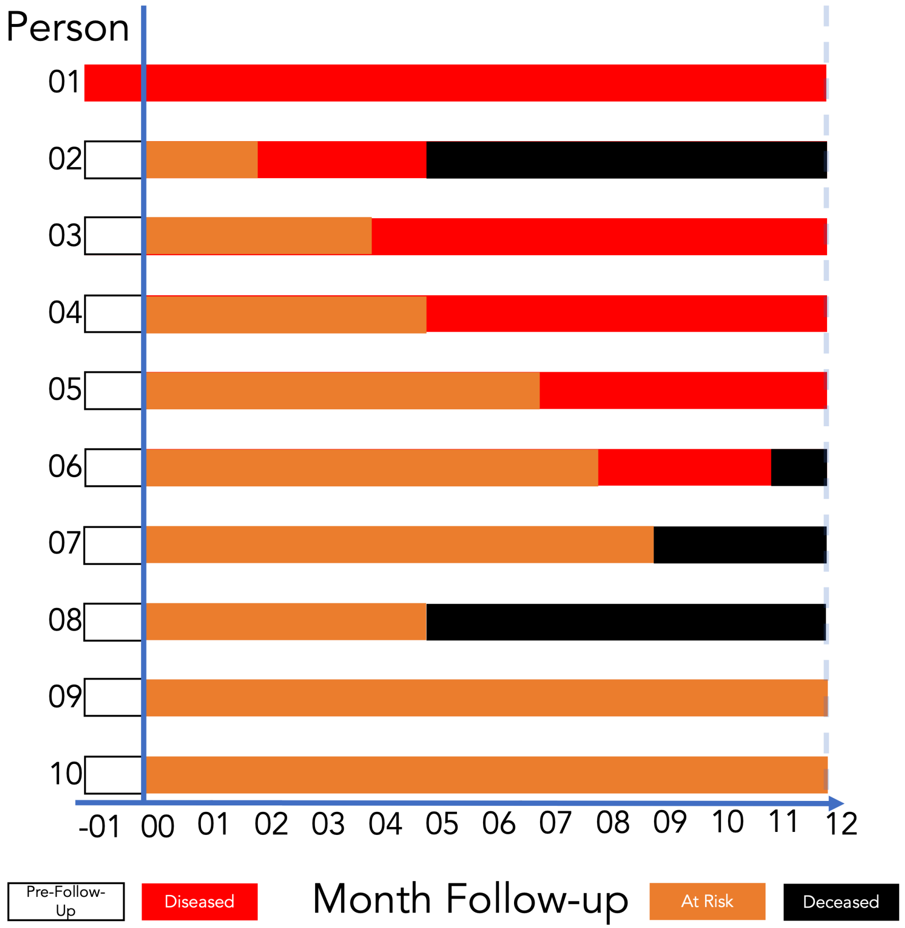
Prevalence Proportion at Month 2



$$\frac{\text{Count with the condition}}{\text{Living members of the population}}$$

$$= \frac{2}{10} = 0.20 = 20\%$$

Interpreting Prevalence Proportion



Among the members of the population, the prevalence of disease in month 2 was **0.20**.

Said another way, **20%** of the population had disease in month 2.

Point Prevalence and Period Prevalence

- **Point prevalence:** prevalence of a condition of interest at a single point in time.
- **Period prevalence:** prevalence of a condition of interest over a period of time.
- As the duration of the period shrinks, period prevalence converges to point prevalence.
- There is no hard-and-fast rule for what window is sufficiently short to count as point prevalence.

Source: (Westreich 2019)

Prevalence Odds

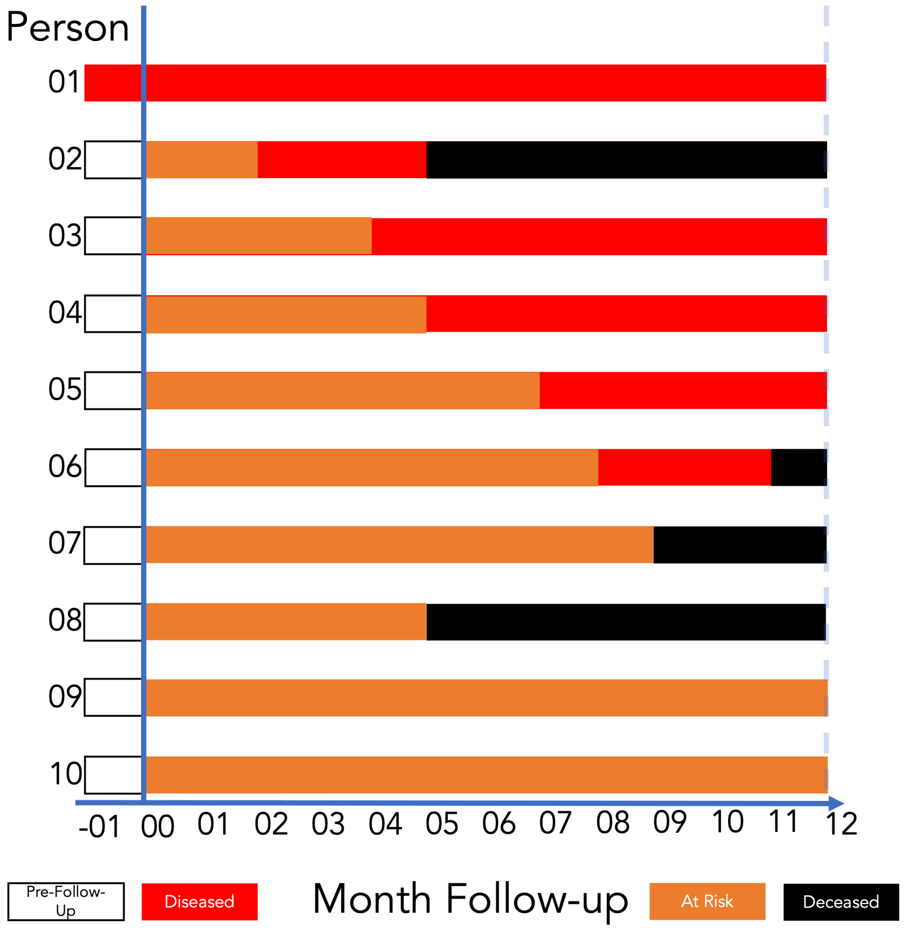
- Prevalence odds is a simple function of prevalence proportion, just as odds in general is a simple function of probability.
- It is typically reported for convenience or desirable statistical properties.
- Prevalence odds is prevalence proportion divided by 1 minus prevalence proportion.
- Range: 0 to infinity.

Source: (Westreich 2019)

Prevalence Odds: Formula

$$\text{Prevalence odds} = \frac{\text{Prevalence proportion}}{1 - \text{Prevalence proportion}}$$

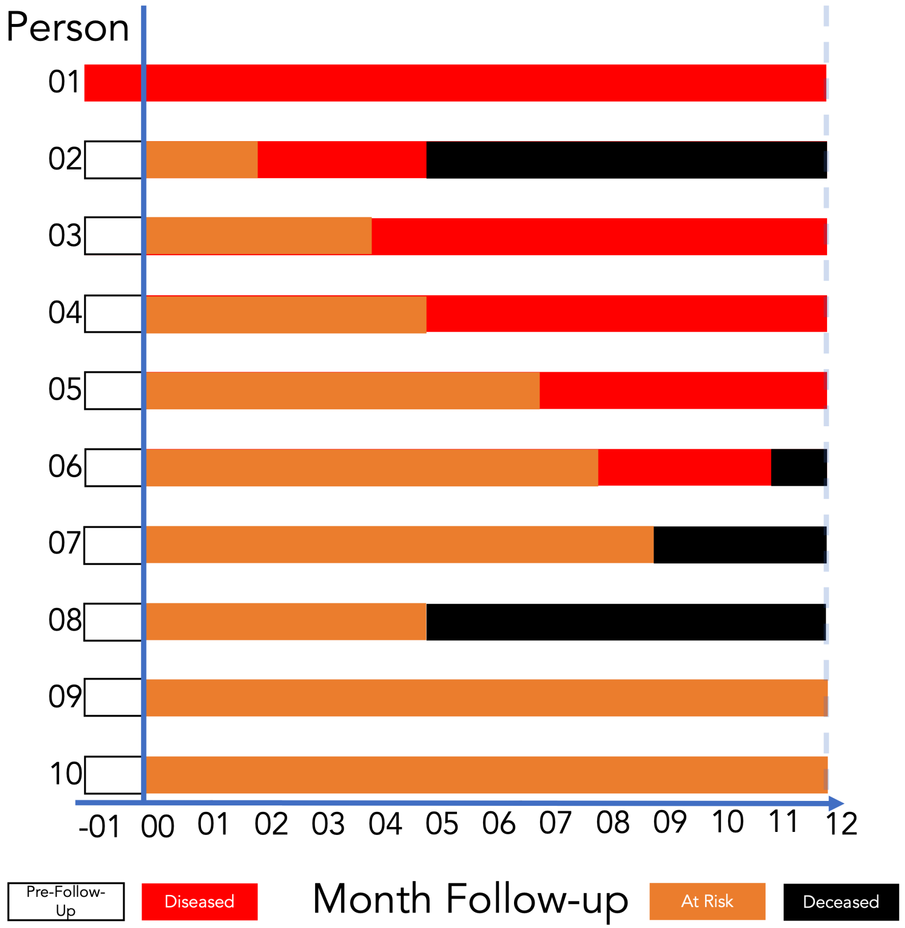
Prevalence Odds at Month 2



$$\frac{\text{Prevalence proportion}}{1 - \text{Prevalence proportion}}$$

$$\text{Prevalence odds} = \frac{0.20}{1 - 0.20} = 0.25$$

Interpreting Prevalence Odds



Among the members of the population, the odds of disease in month 2 were **0.25**.

For every person who had disease at month 2, there were four people who did not.

Incidence Describes New Occurrences

- Prevalence measures how many cases are present at a moment or over a period; incidence measures how many **new cases** arise over a specified period.
- Unlike prevalence, incidence measures occurrences or events: transitions from a condition being absent in an individual to the condition being present over a specified period.

Source: (Westreich 2019)

Incidence Proportion and Risk

- Incidence proportion and “risk” are sometimes used interchangeably in epidemiology.
- Risk can be ambiguous. Incidence proportion is less so.

Source: (Westreich 2019)

Incidence Count

- A count of new occurrences of a condition in a population at risk for the occurrence during a given time frame.
- Range: 0 to infinity.

Source: (Westreich 2019)

Incidence Proportion

- The proportion of the population that experiences a new occurrence of the condition among those at risk of experiencing a new occurrence during a given time frame.
- Like all proportions, it falls between 0 and 1.

Source: (Westreich 2019)

Incidence Proportion: Formula

$$\text{Incidence proportion} = \frac{\text{Count of new occurrences}}{\text{Population at risk}}$$

Follow-Up Experience and New Occurrences

Person	At risk	New disease	Deceased
1	0 months	Prevalent before follow-up	—
2	0–2	Month 2	5–12
3	0–4	Month 4	—
4	0–5	Month 5	—
5	0–7	Month 7	—
6	0–8	Month 8	11–12
7	0–9	—	9–12
8	0–5	—	5–12
9	0–12	—	—
10	0–12	—	—

Month of follow-up. Source: (Lash et al. 2021)

Incidence Proportion Over 12 Months

- 5 people developed disease.
- 9 people were at risk at baseline.
- Person 1 was already diseased and therefore not at risk for a new occurrence.

$$\frac{5}{9} = 0.56 = 56\%$$

Interpreting Incidence Proportion

Among the members of the population, the incidence of disease over 12 months of follow-up was **0.56**.

Fifty-six percent of the population developed disease over 12 months of follow-up.

Incidence Odds: Formula

$$\text{Incidence odds} = \frac{\text{Incidence proportion}}{1 - \text{Incidence proportion}}$$

Incidence Odds Over 12 Months

$$\text{Incidence odds} = \frac{0.56}{1 - 0.56} = 1.27$$

Interpreting Incidence Odds

Among the members of the population, the odds of incident disease over 12 months of follow-up were **1.27**.

For every 127 people who developed incident disease over 12 months of follow-up, 100 people did not.

Incidence Rate

- A measure of the average intensity at which an event occurs in people's experience over time.
- For a population followed from baseline, it is the number of cases divided by the person-time at risk accumulated by the population.
- Range: 0 to infinity.
- **Not a proportion.**
- The denominator is reciprocal time, usually person-time.
- The numeric value depends on the selected time unit and therefore has no interpretation by itself.

Source: (Westreich 2019)

Person-Time

Person-time is the amount of time observed for all people under study while they are at risk of experiencing an incident outcome.

Source: (Westreich 2019)

Follow-Up Experience and Time at Risk

Person	Time at risk	Outcome after risk time
1	0 months	Prevalent disease
2	2 months	Disease, then death
3	4 months	Disease
4	5 months	Disease
5	7 months	Disease
6	8 months	Disease, then death
7	9 months	Death
8	5 months	Death
9	12 months	None observed
10	12 months	None observed

Source: (Lash et al. 2021)

Calculating Person-Time

$$\sum_{\text{people}} \text{Time at risk}$$

$$0 + 2 + 4 + 5 + 7 + 8 + 9 + 5 + 12 + 12 = 64 \text{ person-months}$$

Interpreting Person-Time

The population accumulated **64 person-months at risk** during 12 months of follow-up.

The same amount is **5.33 person-years at risk** during 12 months of follow-up.

Follow-Up Experience for the Incidence Rate

Person	Time at risk	New occurrence?
1	0 months	No; prevalent
2	2 months	Yes
3	4 months	Yes
4	5 months	Yes
5	7 months	Yes
6	8 months	Yes
7	9 months	No
8	5 months	No
9	12 months	No
10	12 months	No

Source: (Lash et al. 2021)

Incidence Rate: Calculation

$$\text{Incidence rate} = \frac{\text{Count of new occurrences}}{\text{Person-time at risk}}$$

$$= \frac{5}{64 \text{ person-months}} \approx 7.8 \text{ per } 100 \text{ person-months}$$

Interpreting the Incidence Rate

The incidence rate of disease among the members of the population was **5 cases per 64 person-months** during 12 months of follow-up.

Equivalently, the incidence rate was **7.8 cases per 100 person-months** during 12 months of follow-up.

References

Lash, Timothy L., Tyler J. VanderWeele, Sebastien Haneuse, and Kenneth J. Rothman. 2021. *Modern Epidemiology*. 4th ed. Wolters Kluwer.

Quinlan, Scott. 2025. *Friis' Epidemiology 101*. 3rd ed. Jones & Bartlett Learning.

Westreich, Daniel. 2019. *Epidemiology by Design: A Causal Approach to the Health Sciences*. Oxford University Press.